

Task 1 — Document 07: Data Flow Diagrams (DFD)

Problem Statement ID: PS 26117 / SIH26117 **Title:** Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work **Organization:** Mangalore Refinery and Petrochemicals Limited (MRPL) **Category:** Software | **Theme:** Smart Automation

1. DFD Level 0 — Context Diagram

The Level 0 Context Diagram represents the system boundary of the **Sovereign AI Workbench**, showing external entities (Plant User, Local Disk Storage, Local Model Runtime) and major data inputs/outputs.

Visual Diagram Flow: GRAPH LR	
Component / Module	Plant Engineer / User
Component / Module	(Local Disk Storage data/)
Component / Module	Local Ollama Model Runtime
User	→ Uploaded Files: PDF, PNG, TXT System
User	→ Chat Queries & Settings System
System	→ Grounded Answers & Code User
System	→ Source Citations & Scores User
System	→ Store Raw Files & JSON Index Storage
Storage	→ Load Vector Index & Chunks System
System	→ Prompt & Image Payloads OllamaRuntime
OllamaRuntime	→ OCR Text, Embeddings, LLM Answers System

2. DFD Level 1 — System Process Overview

Level 1 decomposes the system into four main logical data processes: Document Parsing/OCR, Vector Embedding & Indexing, Task Routing & Retrieval, and Grounded Inference.

Visual Diagram Flow: GRAPH TD	
Component / Module	Plant User
Component / Module	"D1: Knowledge Base Files data/knowledge_base/"
Component / Module	"D2: Vector Index Store data/kb_index.json"
Component / Module	Local Ollama Models
User	→ Upload PDF / Image P1
P1	→ Save Raw File D1
P1	→ Image Payload Ollama
Ollama	→ Extracted OCR Text P1

P1	→ Clean Document Text P2
P2	→ Text Snippets Ollama
Ollama	→ Dense Vector Arrays P2
P2	→ Save Chunks & Vectors D2
User	→ Submit User Prompt P3
P3	→ Read Vectors & Chunks D2
P3	→ Top-5 Matching Context P4

■ 3. DFD Level 2 — Process 1.0 (Multimodal Document & Image Ingestion Pipeline)

Level 2 zooms into Process 1.0 to depict the exact transformation steps from raw uploaded documents to clean text chunks.

■ Visual Diagram Flow: GRAPH TD	
Component / Module	Raw Upload File
Component / Module	1.1 PDF Parser (pypdf Reader)
Component / Module	1.2 OCR Dispatcher (GLM-OCR)
Component / Module	1.3 Text Reader (Direct UTF-8 Read)
Component / Module	1.4 Text Normalizer (Strip outer code fences)
Component / Module	1.5 Semantic Chunker (500 char window / 50 overlap)
Component / Module	Clean Chunk Array to Process 2.0
FileInput	→ FormatCheck
FormatCheck	→ .pdf PDFHandler
FormatCheck	→ .png, .jpg, .webp ImageHandler
FormatCheck	→ .txt, .md TextHandler
PDFHandler	→ Extracted Text CleanText
ImageHandler	→ Extracted Text CleanText
TextHandler	→ Raw Content CleanText
CleanText	→ Chunker

■ 4. Data Dictionary

Data Element	Type	Source	Destination	Description
uploaded_file	Binary Stream	User UI	Process 1.0	PDF document or image file uploaded via form payload.

ocr_prompt_payload	Base64 / Path	Process 1.2	Ollama (glm-ocr)	Image payload sent to local OCR model.
raw_text_stream	String	Process 1.0	Process 1.4	Extracted text before normalization.
chunk_record	JSON Object	Process 2.0	D2: kb_index.json	Contains chunk_id , text , source_file , and 768-dim embedding array.
user_query	String	User UI	Process 3.0	Text prompt entered by user in Web UI.
query_vector	Array[768]	Process 3.0	Cosine Search Engine	768-dimensional float embedding of user query.
cosine_match_score	Float [0.0 - 1.0]	Cosine Search Engine	Process 4.0	Cosine similarity score between query vector and chunk vector.
grounded_response	JSON Object	Process 4.0	User UI	Contains generated LLM answer text, route used, and source document metadata array.